Nonlinear Dynamics in Computer Graphics: Reaction–Diffusion Systems in Procedural Generation

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Abstract

[TODO: Abstract goes here.]

1 Introduction

[TODO: Hook: where do textures in games/films come from? Hand-painted vs. procedural.]

[TODO: Introduce the idea that simple chemical rules can produce animal-coat patterns (Turing 1952), and that this was brought into graphics in 1991 (Turk 1991; Witkin and Kass 1991).]

[TODO: State the aim of this report in one or two sentences, and give a roadmap: Section 2 covers …, Section 3 …, Section 4 …]

2 Background

2.1 Reaction–diffusion systems

[TODO: General form, then the specific model you choose.]

A two-species reaction–diffusion system has the form

$$\frac{\partial u}{\partial t} = f(u,v) + D_u \nabla^2 u, \qquad \frac{\partial v}{\partial t} = g(u,v) + D_v \nabla^2 v, \tag{2.1}$$

where f and g describe the local chemistry and the ∇^2 terms describe diffusion.

3 Undecided

[TODO: Todo 1]

[TODO: Todo 2]

[TODO: Todo 3]

4 Undecided

[TODO: Todo 1]
[TODO: Todo 2]
[TODO: Todo 3]

5 Conclusions

[TODO: Write conclusion]

[TODO: Further directions: e.g. textures on curved surfaces (Turk 1991), GPU/real-time implementation, other models.]

References

Turing, Alan M. (1952). "The Chemical Basis of Morphogenesis". In: *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences* 237.641, pp. 37–72. DOI: 10.1098/rstb.1952.0012.

Turk, Greg (1991). "Generating Textures on Arbitrary Surfaces Using Reaction-Diffusion". In: *Computer Graphics (Proceedings of SIGGRAPH '91)* 25.4, pp. 289–298. DOI: 10.1145/127719.122749.

Witkin, Andrew and Michael Kass (1991). "Reaction-Diffusion Textures". In: *Computer Graphics (Proceedings of SIGGRAPH '91)* 25.4, pp. 299–308. DOI: 10.1145/127719.122750.